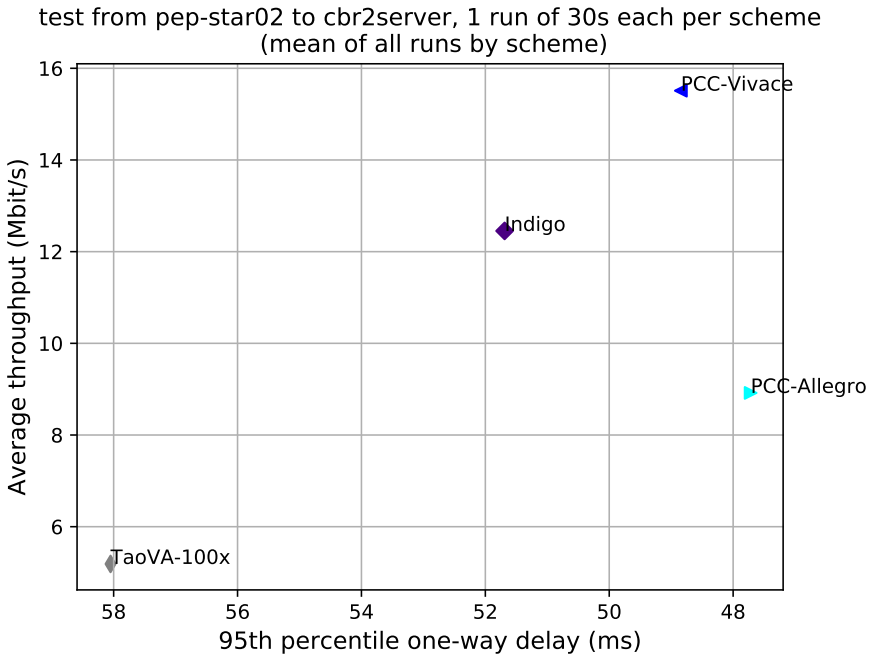
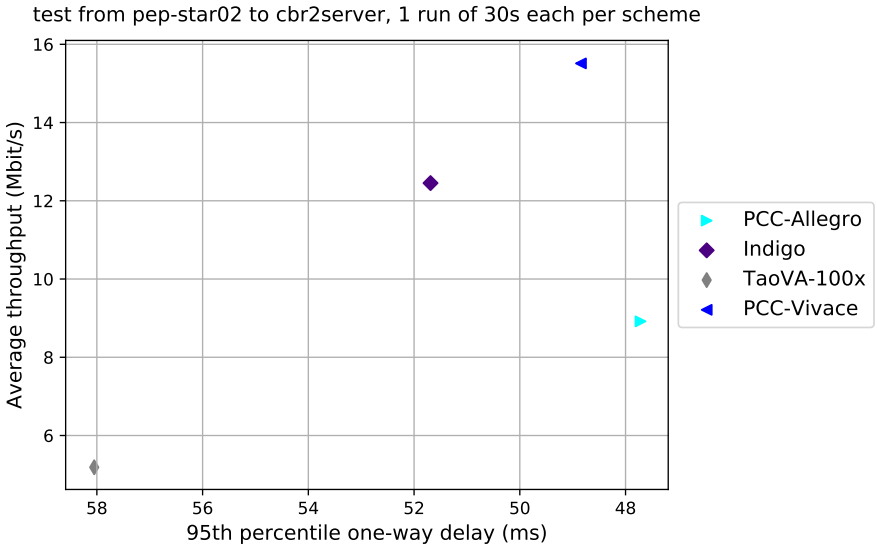
Pantheon Report

Generated at 2026-11-09 23:39:45 (UTC).
Data path: pep-star02 (*local*) →cbr2server (*remote*).
Repeated the test of 4 congestion control schemes once.
Each test lasted for 30 seconds running 1 flow.
NTP offsets were measured against `202.65.114.202` and have been applied to correct the timestamps in logs.

```
System info:
Linux 7.0.0-42-generic
net.core.default_qdisc = fq
net.core.rmem_default = 212992
net.core.rmem_max = 4194304
net.core.wmem_default = 212992
net.core.wmem_max = 4194304
net.ipv4.tcp_rmem = 4096 131072 33554432
net.ipv4.tcp_wmem = 4096 16384 4194304

Git summary:
branch: HEAD @ 23e738ce5acae1d36e321886cd613b0b9401ac11
third_party/genericCC @ d0153f8e594aa89e93b032143cedbdfe58e562f4
third_party/indigo @ 463d89b09699a57bfdfbae351646df6a60040b90
third_party/pantheon-tunnel @ f866d3f58d27afd942717625ee3a354cc2e802bd
third_party/pcc @ 1afc958fa0d66d18b623c091a55fec872b4981e1
 M receiver/src/buffer.h
 M receiver/src/core.cpp
 M sender/src/buffer.h
 M sender/src/core.cpp
third_party/vivace @ 2baf86211435ae071a32f96b7d8c504587f5d7f4
```


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
(mean of all runs by scheme)
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Vivace
Indigo
PCC-Allegro
TaoVA-100x
16
14
12
10
8
6
58
56
54
52
50
48


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Allegro
Indigo
TaoVA-100x
PCC-Vivace
16
14
12
10
8
6
58
56
54
52
50
48

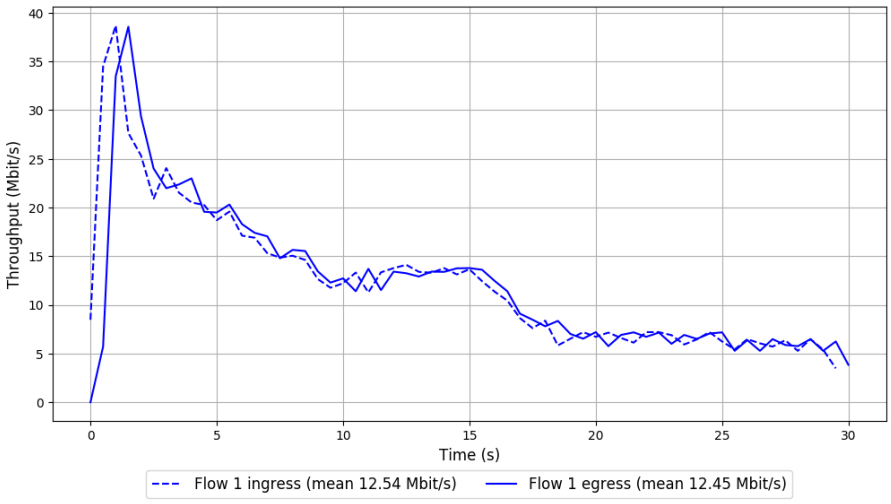
scheme	# runs	mean avg tput (Mbit/s) flow 1	mean 95th-%ile delay (ms) flow 1	mean loss rate (%) flow 1
Indigo	1	12.45	51.69	0.64
PCC-Allegro	1	8.92	47.72	0.02
TaoVA-100x	1	5.19	58.05	1.03
PCC-Vivace	1	15.51	48.84	2.82

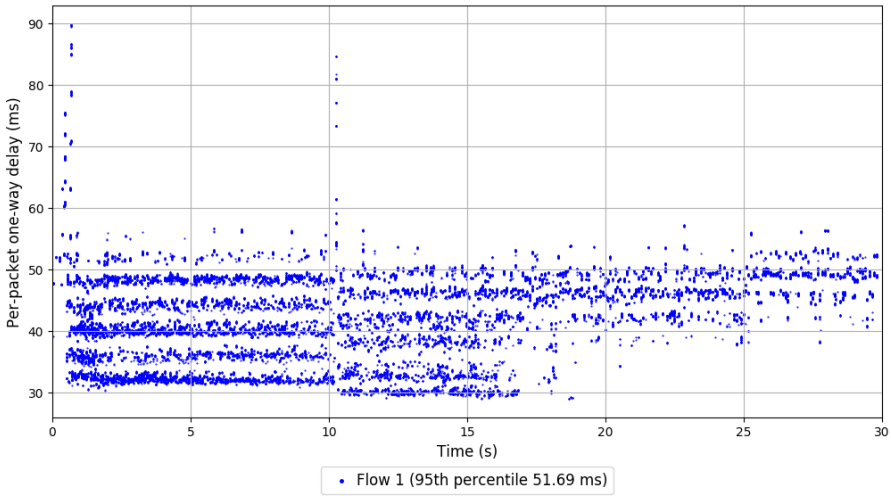
```
Run 1: Statistics of Indigo

Start at: 2026-11-09 22:21:46
End at: 2026-11-09 22:22:16
Local clock offset: 1.757 ms
Remote clock offset: -0.641 ms

# Below is generated by plot.py at 2026-11-09 23:39:44
# Datalink statistics
-- Total of 1 flow:
Average throughput: 12.45 Mbit/s
95th percentile per-packet one-way delay: 51.689 ms
Loss rate: 0.64%
-- Flow 1:
Average throughput: 12.45 Mbit/s
95th percentile per-packet one-way delay: 51.689 ms
Loss rate: 0.64%
```

Run 1: Report of Indigo — Data Link



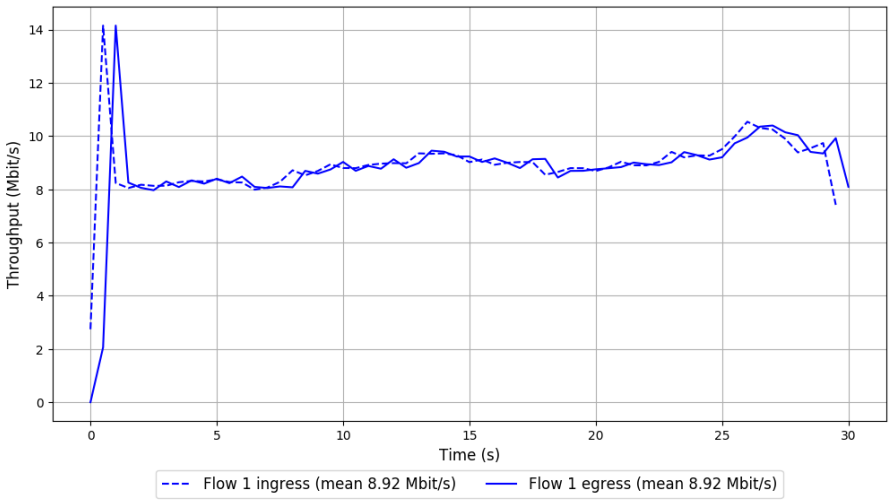


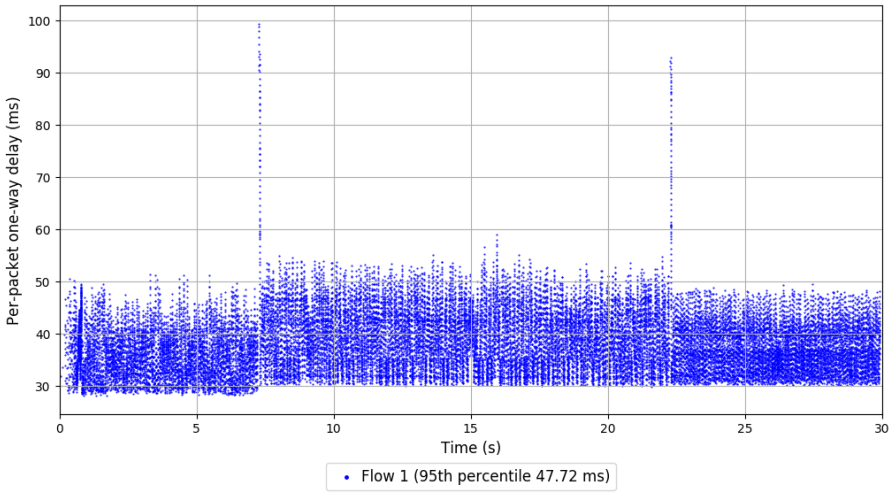
```
Run 1: Statistics of PCC-Allegro

Start at: 2026-11-09 22:24:04
End at: 2026-11-09 22:24:34
Local clock offset: 1.829 ms
Remote clock offset: -0.411 ms

# Below is generated by plot.py at 2026-11-09 23:39:44
# Datalink statistics
-- Total of 1 flow:
Average throughput: 8.92 Mbit/s
95th percentile per-packet one-way delay: 47.717 ms
Loss rate: 0.02%
-- Flow 1:
Average throughput: 8.92 Mbit/s
95th percentile per-packet one-way delay: 47.717 ms
Loss rate: 0.02%
```

Run 1: Report of PCC-Allegro — Data Link



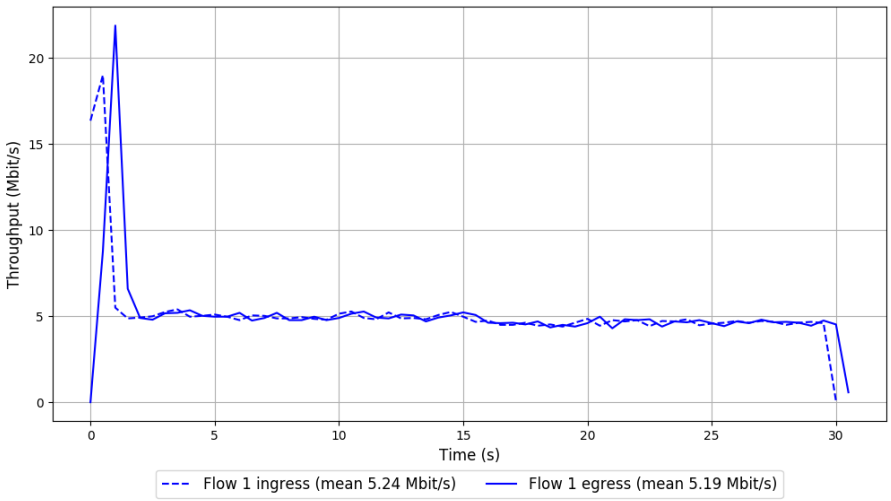


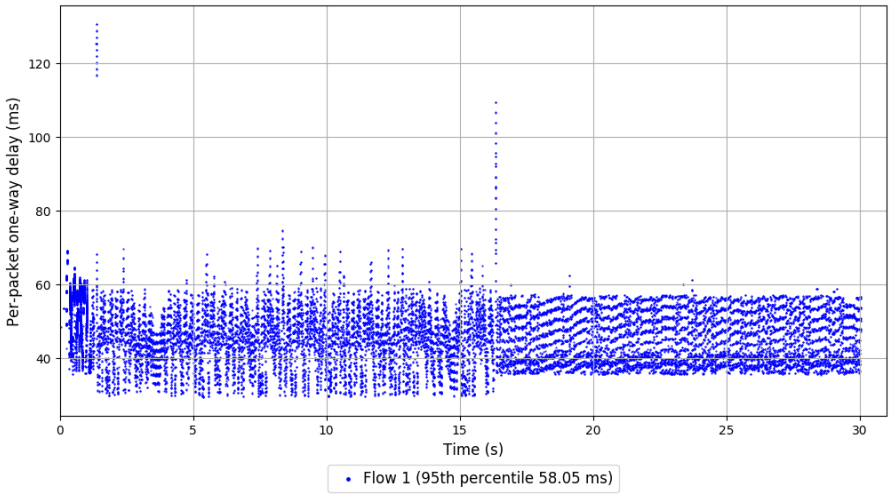
```
Run 1: Statistics of TaoVA-100x

Start at: 2026-11-09 22:22:55
End at: 2026-11-09 22:23:25
Local clock offset: 1.886 ms
Remote clock offset: 0.606 ms

# Below is generated by plot.py at 2026-11-09 23:39:44
# Datalink statistics
-- Total of 1 flow:
Average throughput: 5.19 Mbit/s
95th percentile per-packet one-way delay: 58.050 ms
Loss rate: 1.03%
-- Flow 1:
Average throughput: 5.19 Mbit/s
95th percentile per-packet one-way delay: 58.050 ms
Loss rate: 1.03%
```

Run 1: Report of TaoVA-100x — Data Link





```
Run 1: Statistics of PCC-Vivace

Start at: 2026-11-09 22:20:37
End at: 2026-11-09 22:21:07
Local clock offset: 1.812 ms
Remote clock offset: 0.19 ms

# Below is generated by plot.py at 2026-11-09 23:39:45
# Datalink statistics
-- Total of 1 flow:
Average throughput: 15.51 Mbit/s
95th percentile per-packet one-way delay: 48.844 ms
Loss rate: 2.82%
-- Flow 1:
Average throughput: 15.51 Mbit/s
95th percentile per-packet one-way delay: 48.844 ms
Loss rate: 2.82%
```

Run 1: Report of PCC-Vivace — Data Link

